

profiling terminology / tools

python -m cprofile -o profile.prof
-m sphinx -b html, build/html

↓
binary file
storing all the
profiling data.

ncalls → no. of calls

tottime → time spent inside the function
itself, excluding any funcs it calls

cumtime → time spent inside func + everything
it calls.

percall → $\text{tottime} / \text{ncalls}$

→ $\text{cumtime} / \text{primitive calls}$

gprof 2 dot → gives a call hierarchy

↳ the edges' weights increase
as more cumulative time flows
through them.

Viz Tracer

sphinx-build → comes with sphinx
→ launcher script created
by pip

→ `'viztracer sphinx-build -b html . build'`
→ gives `result.json`
↓
`'vizviewer result.json'`

Learnings & Observations

Some idea about different stages of build could be observed through the profiling outputs. My understanding of the build pipeline based on looking at different profiles: (so far)

Builder.build()

→ Reading stage

- Builder.read()

- -read-serial()

- read-doc()

- io.read() → reads .rst files

- parser.parse() → parse reST

- apply-transform() → something

- store doctree? publish? to do with document tree?

→ Writing stage

- Builder.write()

- write-documents()

- -write-serial()

- get-and-resolve-doctree() - ?

- cross-reference toctrees?

- apply-post-transform()

- write-doc()

- HTMLWriter.translate() - ?



Things I don't understand:

→ transforms?

→ dispatch - visit?

→ Search Indexer feed()

→ previous page: things with
"?" beside them.

from sphinx website (Build phases)

Initialization → Reading → Consistency
checks



Writing ← Resolving